Pandemic Proximity

Did COVID-19 Redraw the Geography of Korea's National R&D Collaboration?

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haseon-jeong.github.io/bio/appam2026

SHORT ANSWER **No clear jump in 2020, but distance trends for new partnerships steepened afterward, including for universities and SMEs — a pandemic-era acceleration.**

Why it matters

Proximity has long been treated as a precondition for R&D collaboration. If remote collaboration became routine during the pandemic, the distance constraint on choosing partners may have loosened — but **for whom**?

- **RQ1** Did partners chosen after 2020 sit farther away?
- **RQ2** Which institutions and research stages drove the change?
- **RQ3** What structurally shapes collaboration distance?

Data

Korea's National R&D project micro-data (NTIS), 2015–2024: lead institution ↔ domestic partner, **route (road) distance** in km.

131,813 new-tie records (main unit)	**65,602** projects
292,966 all tie-year records	**10 yrs** 2015–2024

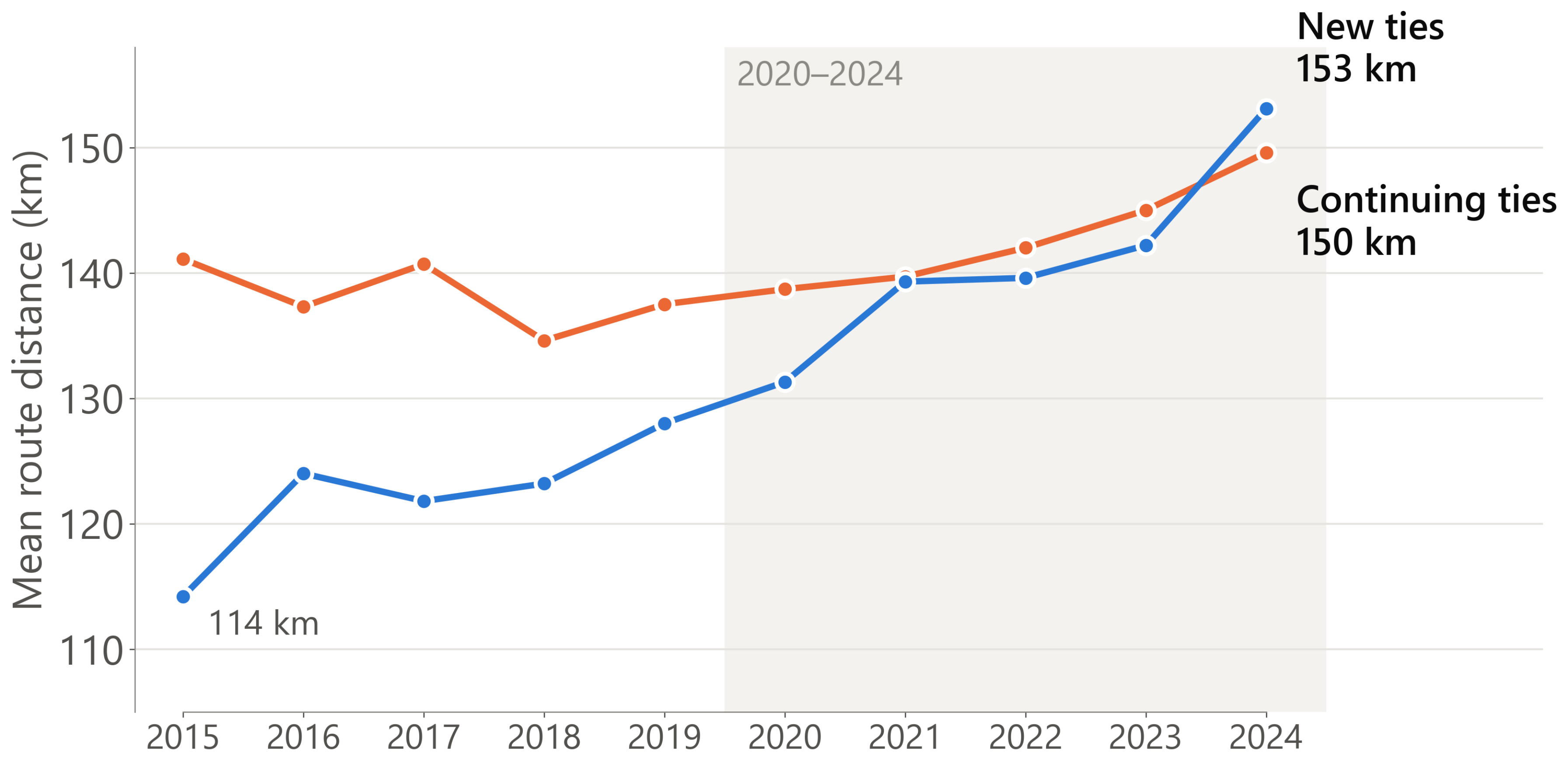
Why new ties? Of 77,200 multi-year ties, only **5.7%** ever change distance. A tie's distance is fixed when the partner is chosen, so we focus on **partner selection**. A new tie is a project × partner pair in its first year (history from 2012).

Design

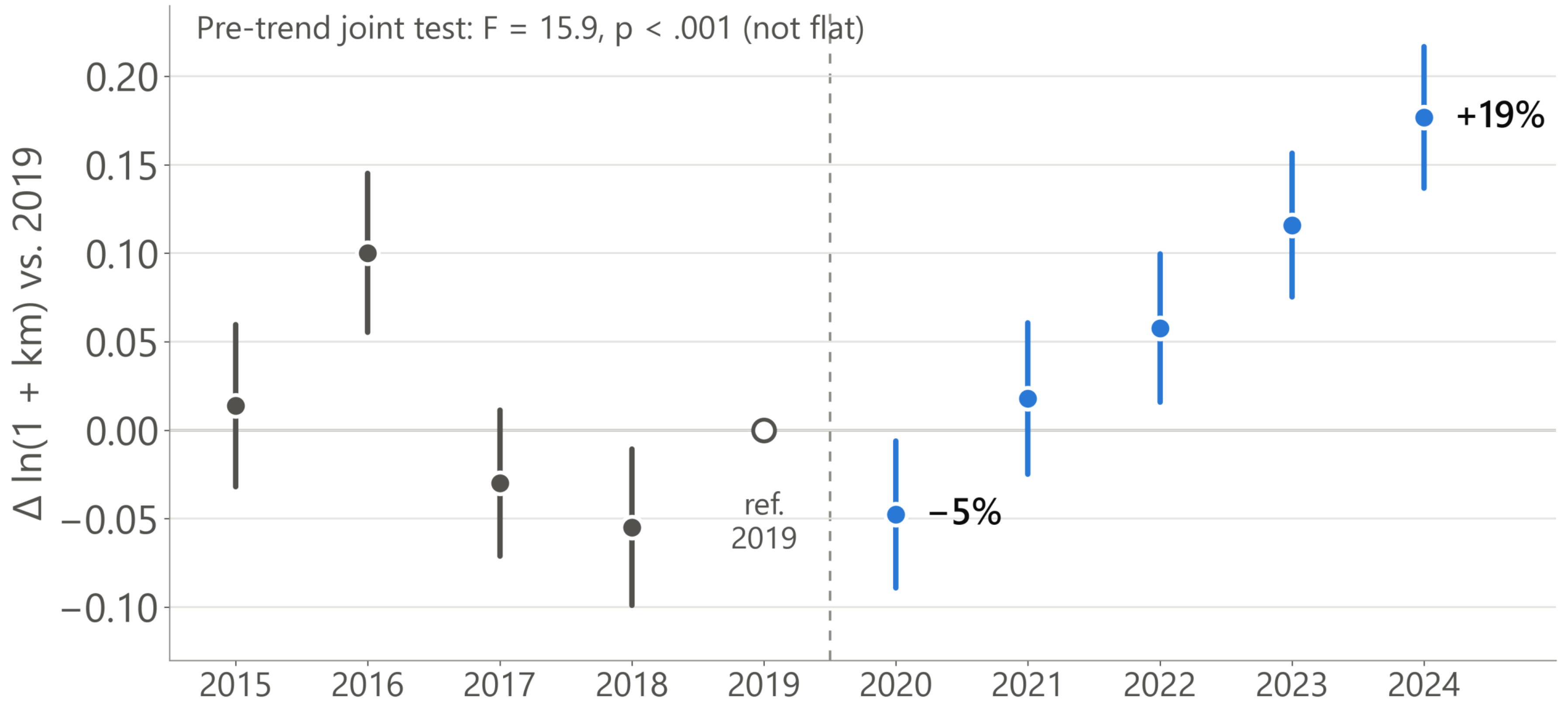
$$\ln(1+\mathrm{km})_{ijt} = \beta_1 \mathrm{Year}_t + \beta_2 \mathrm{Post}_t + \beta_3 (\mathrm{Year}_t - 2020)\cdot \mathrm{Post}_t + X_{ijt}\gamma + \mu_{\mathrm{region}} + \varepsilon$$

- Interrupted time series: β_2 = 2020 departure from pre-trend, β_3 = slope change
- Event study, ref. 2019; SEs clustered by project
- X: institution type, research stage, ln(budget), private funding share; region FE

1 No clear 2020 jump — the trend steepened afterward



Mean route distance of newly formed vs. continuing ties, 2015–2024.

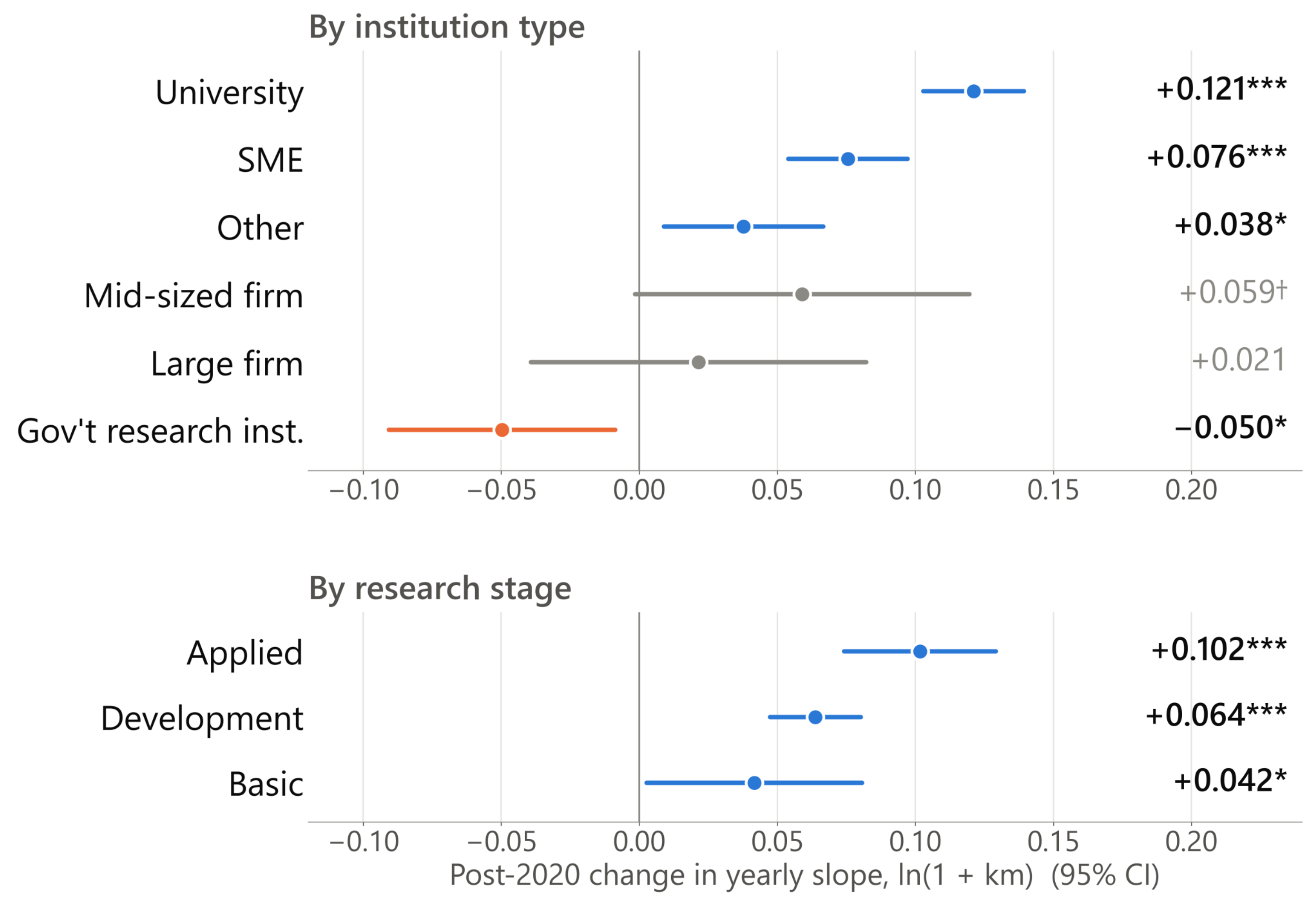


Event study, new ties, with controls (ref. 2019, 95% CI).

2020 vs. pre-trend: +0.006 (n.s.) · **yearly slope, ln(1 + km):** −0.018 → +0.055 (Δ **+0.073*****) · **2024 vs. 2019: ≈ +19%** (event study).

Not a causal estimate: pre-2020 years differ from 2019 (joint F = 15.9, p < .001). Inter-regional share of new ties rose 69% → 75% in 2020, but not after controls (−2 pp) — consistent with a shift in *which* projects started.

2 Trend slopes shifted upward for universities and SMEs and downward for GRIs; the change for large firms was inconclusive



Change in yearly slope after 2020 (not level), new ties; pooled model with group interactions. Blue/orange: p < .05; gray: n.s.; † p < .1.

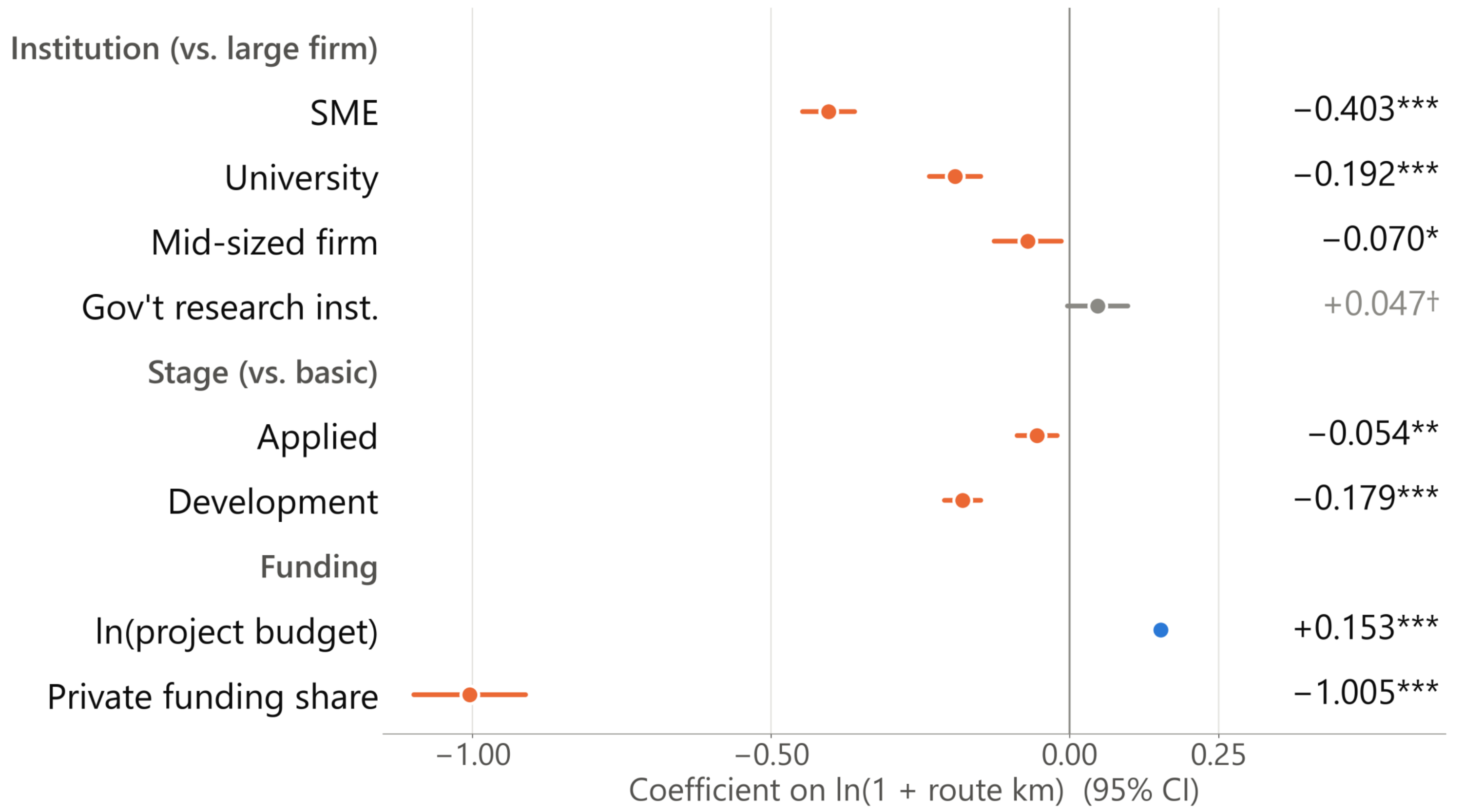
Mean km, new ties	2015–19	2020–24
SME	104.7	123.8
University	121.5	152.9
Gov't research inst.	173.6	153.8
Large firm	165.6	166.0

Mean distances of short- and long-reach institutions **converged**. Mean distance rose at every stage; the slope increase was largest for **applied research** (vs. basic p = .013, vs. development p = .017).

Robustness — post-2020 slope change

Specification	β_3 / yr
New ties, controls (main)	+0.073***
New ties, no controls	+0.029***
Official new-project flag (excl. 2020)	+0.069***
Excluding 2024	+0.073***
Quadratic time trend	+0.065*
Break year 2021 (instead of 2020)	+0.071***

3 Gaps persist across institutions and stages



Level differences (not changes): OLS on new ties with year FE, N = 130,890 (923 missing budget or region), R^2 = 0.09.

SMEs remain the most local (≈ 33% shorter than large firms, conditional on controls), and distance falls from basic to development research. Bigger budgets go with farther partners; private co-funding with closer ones.

Takeaways

1. **Distance gaps across institution types narrowed:** after 2020, distances to new partners rose for **SMEs and universities**, the institutions with the shortest reach before 2020. Whether this widened *access* is the next question.
2. **Gaps persist:** SMEs and development-stage projects remain the most local.
3. **Policy options to evaluate:** matching intermediaries for SMEs, minimum-scale consortia, stage-differentiated incentives for development projects.

Limitations. Pre-trends are not flat (F = 15.9), so we report a *pandemic-era acceleration* — a change observed in that period, not a causal effect. Placebo breaks within 2015–19 (2017, 2018; new ties) are also statistically significant, limiting the interpretation of the estimated trend change as specific to the pandemic period. 2022–24 overlaps other R&D policy changes (e.g., the 2024 budget restructuring).

Data: NTIS National R&D project micro-data, 2015–2024. Also positive: all tie-year records +0.052***; inter-regional tie share (LPM) +0.022***.

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